Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

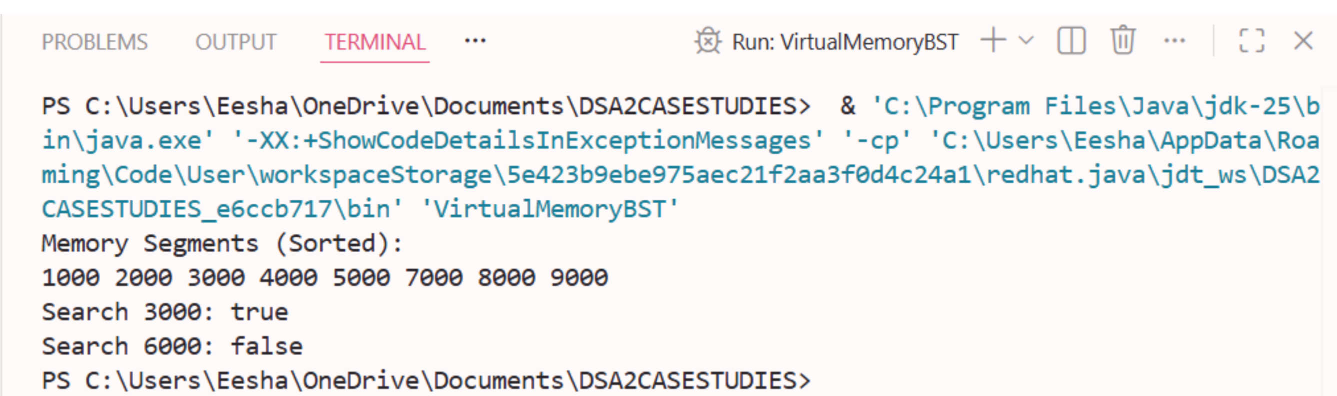
Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	TANGUTURU MADHURI
Student Name	GOTTIPATI EESHA YUKTHA
Roll Number	2520030436
Branch	CSE
Course Outcome	CO1

Case Study Topic: Virtual Memory Management

Work Done Summary:
In this case study, we implemented a Binary Search Tree (BST) to simulate Virtual Memory Management, inspired by how the Linux Kernel uses Red-Black Trees (a self-balancing BST variant) to manage virtual memory segments.
The BST stores memory segment addresses in sorted order so that the OS can quickly allocate, search, and free memory regions. We inserted 8 memory addresses into the BST and observed in-order traversal, search, and deletion operations.
We also implemented a function to find the best-fit free memory segment (closest address >= requested) using BST properties.

Implementation Details:
1. Inserted memory segment base addresses into BST in arrival order.
2. Maintained BST ordering: left child < parent < right child.
3. Performed in-order traversal to display sorted memory layout.
4. Implemented search to check if a segment is allocated.
5. Implemented deletion to simulate memory deallocation.
6. Concluded that BST allows O(log n) allocation and search, suitable for OS memory managers.

Result: Screenshots / Output


```
PROBLEMS   OUTPUT   TERMINAL   ...   Run: VirtualMemoryBST

PS C:\Users\Eesha\OneDrive\Documents\DSA2CASESTUDIES>  & 'C:\Program Files\Java\jdk-25\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Eesha\AppData\Roaming\Code\User\workspaceStorage\5e423b9ebe975aec21f2aa3f0d4c24a1\redhat.java\jdt_ws\DSA2CASESTUDIES_e6ccb717\bin' 'VirtualMemoryBST'
Memory Segments (Sorted):
1000 2000 3000 4000 5000 7000 8000 9000
Search 3000: true
Search 6000: false
PS C:\Users\Eesha\OneDrive\Documents\DSA2CASESTUDIES>
```

- BST provided O(log n) insertion and search for memory segment addresses.
- In-order traversal returned sorted memory layout: 1000, 2000, 3000, 4000, 5000, 7000, 8000, 9000.
- Search for 3000: true | Search for 6000: false (not allocated).
  Time Complexity:
- Insert / Search / Delete: O(log n) average, O(n) worst case.
- Compared to a simple array, BST performs significantly better for dynamic OS memory operations due to its hierarchical sorted structure.

GitHub Link:
https://github.com/egotpat-gif/DSAII-Case-Studies/tree/main/DSA_CO1_BST_CASESTUDY

Student Signature	Faculty Signature